

Theory and Algorithm Notes for MieScat_Py

Lorenz–Mie Scattering by Neutral and Charged Homogeneous Spheres

Technical Note for MieScat_Py v0.2.0

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1 Introduction

`MieScat.Py` solves the classical problem of electromagnetic scattering by a single homogeneous isotropic sphere embedded in a nonabsorbing medium. The implementation follows the formulations in Bohren and Huffman [1] (hereafter BH83) and Mishchenko et al. [2], with an extension to electrically charged spheres based on Klacka and Kocifaj [3] (hereafter KK10). The forward and backward complex scattering amplitudes used for Complex Amplitude Sensing (CAS) are defined according to Moteki [4] and Moteki and Adachi [5].

This note documents the mathematical formulations, numerical algorithms, and their correspondence to the code. Sec. 2 covers the standard Mie theory for neutral spheres, from the problem definition through to the computed observables. Sec. 3 describes the extension to charged spheres. Sec. 4 defines the CAS-specific observables. Sec. 5 summarizes the computational implementation.

2 Mie theory for neutral spheres

2.1 Problem definition and notation

Consider a sphere of diameter d_p (radius $r_p = d_p/2$) with complex refractive index

$$m_p = n_p + i \kappa_p \quad (1)$$

embedded in a nonabsorbing medium with real refractive index m_m . A monochromatic plane wave of vacuum wavelength λ_0 illuminates the sphere. The vacuum wavenumber, medium wavenumber, size parameter, and relative refractive index are

$$k_0 = \frac{2\pi}{\lambda_0}, \quad k = m_m k_0, \quad x = k r_p = \frac{\pi d_p m_m}{\lambda_0}, \quad m_r = \frac{m_p}{m_m}. \quad (2)$$

The number of terms retained in the partial wave expansion is

$$N_{\text{stop}} = \lfloor |x| + 4|x|^{1/3} + 2 \rfloor, \quad (3)$$

following the criterion of BH83.

2.2 Ricatti–Bessel functions and logarithmic derivative

The Mie solution is expressed in terms of the Ricatti–Bessel functions

$$\psi_n(z) = z j_n(z), \quad \chi_n(z) = -z y_n(z), \quad \xi_n(z) = \psi_n(z) + i \chi_n(z) = z h_n^{(1)}(z), \quad (4)$$

where j_n , y_n , and $h_n^{(1)}$ are the spherical Bessel, Neumann, and Hankel functions of the first kind, respectively.

The logarithmic derivative of ψ_n at argument $y = m_r x$ is

$$D_n(y) = \frac{d}{dy} \ln \psi_n(y) = \frac{\psi_n'(y)}{\psi_n(y)}. \quad (5)$$

It is computed by the numerically stable *downward* recurrence (BH83, Sec. 4.8)

$$D_{n-1}(y) = \frac{n}{y} - \frac{1}{D_n(y) + n/y}, \quad n = N_{\text{mx}}, N_{\text{mx}} - 1, \dots, 1, \quad (6)$$

starting from $D_{N_{\text{mx}}}(y) = 0$, where $N_{\text{mx}} = \lfloor \max(N_{\text{stop}}, |y|) + 15 \rfloor$.

$\psi_n(x)$ is computed via the ratio $R_n = \psi_n(x)/\psi_{n-1}(x)$ using downward recurrence (Mishchenko et al. [2]):

$$R_n = \frac{1}{(2n+1)/x - R_{n+1}}, \quad n = N_{\text{mx}} - 1, \dots, 0, \quad R_{N_{\text{mx}}} = \frac{x}{2N_{\text{mx}} + 1}, \quad (7)$$

followed by $\psi_0(x) = R_0 \cos x$ and $\psi_n(x) = R_n \psi_{n-1}(x)$ for $n \geq 1$.

$\chi_n(x)$ is computed by upward recurrence, which is stable for this function:

$$\chi_0(x) = -\cos x, \quad \chi_1(x) = \frac{\chi_0(x)}{x} - \sin x, \quad \chi_n(x) = \frac{2n-1}{x} \chi_{n-1}(x) - \chi_{n-2}(x). \quad (8)$$

2.3 Partial wave coefficients

For a neutral (uncharged) sphere, the partial wave (Mie) coefficients a_n and b_n are (BH83, Eq. 4.88)

$$a_n = \frac{(D_n(y)/m_r + n/x) \psi_n(x) - \psi_{n-1}(x)}{(D_n(y)/m_r + n/x) \xi_n(x) - \xi_{n-1}(x)}, \quad (9)$$

$$b_n = \frac{(m_r D_n(y) + n/x) \psi_n(x) - \psi_{n-1}(x)}{(m_r D_n(y) + n/x) \xi_n(x) - \xi_{n-1}(x)}, \quad (10)$$

where $y = m_r x$ and $n = 1, 2, \dots, N_{\text{stop}}$. The coefficients a_n correspond to the electric (TM) multipoles and b_n to the magnetic (TE) multipoles.

2.4 Optical efficiency factors

Given a_n and b_n , the optical efficiency factors are (BH83, Eqs. 4.61–4.62)

$$Q_{\text{sca}} = \frac{2}{x^2} \sum_{n=1}^{N_{\text{stop}}} (2n+1) (|a_n|^2 + |b_n|^2), \quad (11)$$

$$Q_{\text{ext}} = \frac{2}{x^2} \sum_{n=1}^{N_{\text{stop}}} (2n+1) \text{Re}(a_n + b_n), \quad (12)$$

$$Q_{\text{abs}} = Q_{\text{ext}} - Q_{\text{sca}}, \quad (13)$$

$$Q_{\text{back}} = \frac{1}{x^2} \left| \sum_{n=1}^{N_{\text{stop}}} (2n+1) (-1)^n (a_n - b_n) \right|^2. \quad (14)$$

These are dimensionless ratios of the respective cross sections to the geometric cross section πr_p^2 .

2.5 Scattering amplitudes

The angle-dependent functions $\pi_n(\cos \theta)$ and $\tau_n(\cos \theta)$ are defined as

$$\pi_n(\cos \theta) = \frac{P_n^1(\cos \theta)}{\sin \theta}, \quad \tau_n(\cos \theta) = \frac{dP_n^1(\cos \theta)}{d\theta}, \quad (15)$$

where P_n^1 is the associated Legendre function of degree n and order 1. They satisfy the recurrence relations (with $\mu = \cos \theta$, $\pi_1 = 1$, $\pi_2 = 3\mu$)

$$\pi_n = \frac{2n-1}{n-1} \mu \pi_{n-1} - \frac{n}{n-1} \pi_{n-2}, \quad \tau_n = n \mu \pi_n - (n+1) \pi_{n-1}. \quad (16)$$

The elements of the 2×2 amplitude scattering matrix (BH83, Eq. 4.74) are

$$S_1(\theta) = \sum_{n=1}^{N_{\text{stop}}} \frac{2n+1}{n(n+1)} (a_n \pi_n + b_n \tau_n), \quad (17)$$

$$S_2(\theta) = \sum_{n=1}^{N_{\text{stop}}} \frac{2n+1}{n(n+1)} (a_n \tau_n + b_n \pi_n). \quad (18)$$

The code converts these to the amplitude matrix convention of Mishchenko et al. [2]:

$$S_{11}(\theta) = \frac{S_2(\theta)}{-ik}, \quad S_{22}(\theta) = \frac{S_1(\theta)}{-ik}. \quad (19)$$

Note the index swap: S_{11} is derived from S_2 , and S_{22} from S_1 . Both have the dimension of length.

2.6 Mass-normalized cross sections

The geometric cross section, particle volume, and particle mass are

$$C_{\text{geo}} = \frac{\pi d_p^2}{4}, \quad V_p = \frac{\pi d_p^3}{6}, \quad M_p = \rho_p V_p, \quad (20)$$

where ρ_p is the particle material density. The mass-normalized cross sections are

$$\text{MSC} = \frac{C_{\text{geo}} Q_{\text{sca}}}{M_p}, \quad \text{MEC} = \frac{C_{\text{geo}} Q_{\text{ext}}}{M_p}, \quad \text{MAC} = \frac{C_{\text{geo}} Q_{\text{abs}}}{M_p}. \quad (21)$$

When d_p is given in m and ρ_p in g cm^{-3} , the code internally converts V_p to cm^3 so that M_p is in g and the mass-normalized cross sections are in $\text{m}^2 \text{g}^{-1}$.

3 Extension to charged spheres

For a sphere carrying n_e elementary charges e in vacuum ($m_m = 1$), the formulation of KK10 introduces a charge-dependent G -parameter that modifies a_n and b_n . The total charge is $Q = n_e e$ and the electrostatic surface potential is

$$V_{\text{surf}} = \frac{Q}{4\pi\epsilon_0 r_p}. \quad (22)$$

The quantum scattering rate is estimated as (KK10, Eq. 37)

$$\gamma_s = f_\gamma \frac{k_B T}{\hbar}, \quad (23)$$

where f_γ is a dimensionless correction factor, k_B is the Boltzmann constant, $T = 298 \text{ K}$, and $\hbar$ is the reduced Planck constant. The dimensionless charge parameter is (KK10, Eq. 26)

$$G = \frac{e V_{\text{surf}}}{m_e c^2} \cdot \frac{1}{x} \cdot \frac{1 + i(\gamma_s/\omega)}{1 + (\gamma_s/\omega)^2}, \quad (24)$$

where m_e is the electron mass, c is the speed of light, and $\omega = k c$ is the angular frequency.

The modified partial wave coefficients for the charged sphere are

$$a_n = \frac{[(1 + nG/x) D_n/m_p + n/x] \psi_n - (1 + G D_n/m_p) \psi_{n-1}}{[(1 + nG/x) D_n/m_p + n/x] \xi_n - (1 + G D_n/m_p) \xi_{n-1}}, \quad (25)$$

$$b_n = \frac{[m_p D_n + (n/x)(1 - Gx/n)] \psi_n - \psi_{n-1}}{[m_p D_n + (n/x)(1 - Gx/n)] \xi_n - \xi_{n-1}}, \quad (26)$$

where all Ricatti–Bessel functions are evaluated at x , and $D_n = D_n(m_p x)$ with $m_r = m_p$ (since the medium is vacuum). When $G = 0$ (i.e., $n_e = 0$), (25)–(26) reduce to the standard Mie expressions (9)–(10) with $m_r = m_p$. The efficiency factors, scattering amplitudes, and mass-normalized cross sections are then computed identically to Sec. 2.4–2.6.

4 Complex Amplitude Sensing observables

For the CAS protocol [4, 5], the relevant observables are the forward scattering amplitude $S(0)$ and a backward scattering amplitude S_{bak} .

At $\theta = 0$ the angular functions satisfy $\pi_n(1) = \tau_n(1) = 1$ for all n , so $S_1(0) = S_2(0)$ and hence $S_{11}(0) = S_{22}(0)$. The forward scattering amplitude is

$$S(0) = \frac{S_{11}(0) + S_{22}(0)}{2} = S_{11}(0). \quad (27)$$

At $\theta = \pi$ the angular functions satisfy $\pi_n(-1) = (-1)^{n+1}$ and $\tau_n(-1) = (-1)^n$, giving $S_1(\pi) = -S_2(\pi)$. The backward scattering amplitude for CAS-v2 is

$$S_{\text{bak}} = \frac{-S_{11}(\pi) + S_{22}(\pi)}{\sqrt{2}}. \quad (28)$$

The code returns $\text{Re}[S(0)]$, $\text{Im}[S(0)]$, $\text{Re}[S_{\text{bak}}]$, and $\text{Im}[S_{\text{bak}}]$, all having the dimension of length.

5 Computational implementation

Table 1: Modules in `MieScat_Py` v0.2.0 and their roles.

Module	Role
<code>_mie_core.py</code>	Shared helper functions ((6)–(19))
<code>miescat.py</code>	Neutral sphere: Q -factors, S_{11} , S_{22} , MSC/MEC/MAC
<code>miescat_charged.py</code>	Charged sphere (KK10): adds V_{surf} , G -parameter
<code>mie_complex_amplitudes.py</code>	CAS observables: $S(0)$ and S_{bak}

The key numerical considerations are as follows. The logarithmic derivative $D_n(y)$ is computed by *downward* recurrence (6), which is numerically stable for complex y (BH83, Sec. 4.8). $\psi_n(x)$ is computed via the ratio method (7) using downward recurrence on the ratios R_n , which avoids the catastrophic cancellation that plagues upward recurrence for large x (Mishchenko et al. [2]). $\chi_n(x)$ is computed by upward recurrence (8), which is stable for this function. The recurrence relations for D_n , R_n , ψ_n , χ_n , π_n , and τ_n remain as sequential loops because each step depends on the previous value.

In v0.2.0, the following element-wise Python loops were replaced by NumPy vectorized operations: (i) the weight arrays $(2n + 1)$, $(2n + 1)/[n(n + 1)]$, and $(-1)^n$; (ii) the partial wave coefficients a_n , b_n ((9)–(10) or (25)–(26)) for all n simultaneously; (iii) the scattering amplitude sums (17)–(18) over n for all angles θ simultaneously via array broadcasting.

Table 2: Correspondence between mathematical symbols and code variables.

Symbol	Code variable	Description
λ_0	<code>wl_0</code>	Vacuum wavelength
m_m	<code>m_m</code>	Medium refractive index
d_p	<code>d_p</code>	Particle diameter
m_p	<code>m_p</code>	Complex particle refractive index
m_r	<code>m_r</code>	Relative refractive index
x	<code>x</code>	Size parameter
N_{stop}	<code>nstop</code>	Number of expansion terms
$D_n(y)$	<code>DD[n]</code>	Logarithmic derivative
$\psi_n(x)$	<code>psi[n]</code>	Ricatti-Bessel (1st kind)
$\chi_n(x)$	<code>chi[n]</code>	Ricatti-Bessel (2nd kind)
$\xi_n(x)$	<code>xi[n]</code>	Ricatti-Bessel (3rd kind)
a_n	<code>a[n]</code>	Electric multipole coefficient
b_n	<code>b[n]</code>	Magnetic multipole coefficient
π_n	<code>pi_n[n,:]</code>	Angular function
τ_n	<code>tau_n[n,:]</code>	Angular function
S_1, S_2	<code>S1, S2</code>	BH83 amplitude elements
S_{11}, S_{22}	<code>S11, S22</code>	Mishchenko amplitude elements
G	<code>g</code>	Charge parameter (KK10)
V_{surf}	<code>surf_potential</code>	Surface potential
ρ_p	<code>dens_p</code>	Particle density

References

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